Prediction of Bank Credit Worthiness Using Machine Learning Algorithms

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ABSTRACT

Predicting credit worthiness of clients in banking is a very important issue. Large datasets with numerous variables must be analyzed for this activity. In this study, real bank credit data was utilized to predict a customer's creditworthiness using three distinct Machine Learning algorithms. We analyze which algorithms perform best for the given dataset. The dataset used for this research is "default of credit card clients" taken from the UCI Machine Learning Repository. There are 23 total features in the dataset. The proportions of clients considered "good" and "bad" are often not balanced. We have applied different techniques like Under-Sampling and Over-Sampling and SMOTE to work with imbalanced data to get the best results for each of the algorithms we have used. Each of the Machine Learning Classifiers we utilize is paired with a variety of feature selection strategies and data balance techniques. After analyzing a dataset from a retail credit bank, we determined that the most effective approach is to use Extra Trees in conjunction with Random Over-Sampling.

Keywords: Creditworthiness, Machine Learning, Credit card, imbalanced data, Under-Sampling, Over-Sampling, SMOTE.

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LIST OF ABBREVIATIONS

ML	Machine Learning
ET	Extra Trees
DT	Decision Tree
LR	Logistic Regression
SMOTE	Synthetic Minority Over-sampling Technique
MARS	Multivariate Adaptive Regression Splines
LDA	Linear Discriminant Analysis
SVM	Support Vector Machines
ANN	Artificial Neural Networks
RF	Random Forest
GB	Gradient Boosting
NN	Neural Networks
CART	Classification and Regression Tree

ASM	Attribute Selection Measure
CV	Cross Validation
CM	Confusion matrix
TP	True Positive
FP	False Positive
TN	True Negative
FN	False Negative
KNN	K-Nearest Neighbors
UCI	University of California Irvine
PCA	Principal Component Analysis
CNN	Convolutional Neural Networks
RNN	Recurrent Neural Networks